

Question 3

A car rental company has a fleet of 1000 cars, and each car has four tires. The company seeks to estimate the proportion of tires that require replacement. The company statistician draws a simple random sample of 10 cars from the fleet, and for each sampled car determines the number of tires that require replacement. The following is the sample data:

Car	1	2	3	4	5	6	7	8	9	10
Number of tires that Require replacement	0	0	4	0	2	0	4	0	0	2

Estimate the proportion of all tires that require replacement, and give the standard error of your estimate.

Question 4

A random sample of 23 college students are picked at random from a large college. The following stem & leaf gives the ages of 23 college students:

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1 | 88899
2 | 011234568
3 | 012347
4 | 038
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- a) compute the median age in sample _____
b) Test at the 0.05 level of significance the null hypothesis that the Population Median age is 39, versus the alternative that it is less than 39.